

Supply Chain & Logistics — Domain Notes for Data Analysts/Scientists

01. Overview — What is this domain?

Supply Chain & Logistics covers the analytical work behind **moving raw materials and finished goods efficiently from source to end customer**. It spans the full flow of physical goods and the information systems that coordinate them. This domain includes:

- **Procurement & Sourcing** – buying raw materials/components from suppliers
- **Inventory Management** – holding the right stock at the right locations
- **Warehousing & Distribution** – storing, picking, packing, and shipping goods
- **Transportation & Logistics** – moving goods via road, rail, sea, or air
- **Last-Mile Delivery** – the final, often costliest leg of delivery to the end customer
- **Demand & Supply Planning (S&OP)** – balancing forecasted demand with production/procurement capacity
- **Reverse Logistics** – managing returns, repairs, and recycling flows
- **Third-Party Logistics (3PL) & Freight Brokerage** – outsourced logistics service providers

Why data analytics/science matters here

1. **Supply chains are complex, multi-echelon networks** — small local decisions (a supplier delay, a warehouse stockout) ripple across the entire network, and analytics helps manage this complexity.
2. **Costs are enormous and often hidden** — transportation, warehousing, and inventory carrying costs are major expense categories where small percentage improvements yield large absolute savings.
3. **Uncertainty is constant** — demand fluctuates, suppliers face disruptions, and transit times vary; probabilistic and forecasting methods are essential to plan under this uncertainty.
4. **Speed and reliability are competitive differentiators** — especially in e-commerce, delivery speed and consistency directly affect customer satisfaction and retention.
5. **Global disruptions have highlighted supply chain fragility** — analytics increasingly supports resilience and risk management, not just cost optimization.

Industry context & key considerations

Consideration	Relevance
Incoterms (International Commercial Terms)	Standardized rules defining responsibility for shipping, insurance, and customs between buyer and seller
Customs & Trade Compliance	Import/export regulations, tariffs, and documentation requirements for cross-border logistics
Environmental Regulations	Emissions standards affecting transportation mode choice and route planning
Labor Regulations	Driver hours-of-service rules, warehouse labor laws affecting scheduling and capacity planning
SCOR Model (Supply Chain Operations Reference)	Widely used industry framework for standardizing supply chain process definitions and metrics

Typical business functions a Data Analyst/Scientist supports

- Demand Planning & Forecasting
- Inventory Optimization
- Transportation & Route Optimization
- Warehouse Operations Analytics
- Supplier Performance & Risk Management
- Network Design & Distribution Planning
- Sales & Operations Planning (S&OP)

Why this domain is attractive for Data Analysts/Scientists

- Rich optimization problems combining statistics, operations research, and business strategy
- Direct, quantifiable impact on cost savings and service levels
- Growing demand due to increased focus on supply chain resilience post-pandemic
- Career path: Supply Chain Analyst → Senior Analyst/Data Scientist → Supply Chain Analytics Manager → Director of Supply Chain Planning/Chief Supply Chain Officer

02. Role of a Data Analyst/Scientist — Day-to-Day & Full Project Lifecycle

A. Day-to-Day Responsibilities (Snapshot)

- Building and refreshing demand forecasts feeding into procurement and production plans
- Monitoring inventory health dashboards (stock levels, aging inventory, stockout risk)
- Analyzing transportation costs and identifying route/carrier optimization opportunities
- Investigating supply chain disruptions (late shipments, supplier quality issues) with root-cause analysis
- Supporting Sales & Operations Planning (S&OP) meetings with data-driven scenario analysis
- Building/maintaining network optimization models for warehouse/distribution center placement
- Tracking supplier performance scorecards

B. End-to-End Project Lifecycle (Example: Multi-Echelon Inventory Optimization)

Step 1: Business Problem Understanding

- Meet with Supply Chain/Operations leadership to understand the pain point.
 - Example: *"We're carrying \$80M in inventory across our distribution network, but still experience stockouts on 8% of SKU-location combinations weekly, while other locations are overstocked. We need to optimize how inventory is allocated across our network of regional distribution centers (DCs) and stores."*
- Translate into an analytical framing:
 - Business ask → *Reduce total inventory investment while improving service levels*
 - Analytical framing → *Build a multi-echelon inventory optimization model to determine optimal stock levels (safety stock, reorder points) at each node of the network (central DC, regional DCs, stores), accounting for demand variability and lead times*
- Define success metrics: target service level (e.g., 95% in-stock rate), inventory reduction target, forecast accuracy improvement

Step 2: Data Collection & Understanding

- Identify data sources: historical sales/demand by SKU-location, current inventory levels across all network nodes, supplier/transportation lead times, storage capacity constraints
- Understand the network structure (how many echelons — e.g., central DC → regional DC → store — and how inventory flows between them)
- Define service level targets by product category (e.g., fast-moving items may warrant higher service levels than slow-moving ones)

Step 3: Data Extraction

- Pull historical demand data at the SKU-location-day/week level from POS/order systems
- Pull current inventory positions and in-transit inventory from the Warehouse Management System (WMS) and ERP
- Pull lead time data — both supplier-to-DC and DC-to-store/DC-to-DC transit times, including variability, not just averages
- Pull cost data: holding cost per unit, stockout cost estimates, transportation costs between nodes

Step 4: Data Cleaning & Preprocessing

- Handle demand data affected by historical stockouts (as in retail — a stockout period shows zero sales, which understates true demand and must be "unconstrained")
- Reconcile SKU identifiers across systems if the same product has different codes in different regions/systems
- Handle missing or inconsistent lead time records (common when supplier data isn't systematically tracked)
- Segment SKUs by demand volume/variability (e.g., ABC-XYZ classification) since different segments require different optimization treatment

Step 5: Exploratory Data Analysis (EDA)

- Analyze demand variability by SKU and location (coefficient of variation is a key input to safety stock calculations)
- Analyze lead time variability from suppliers and between network nodes

- Identify current service level performance and inventory turns by category/location to establish a baseline
- Visualize the network structure and current inventory distribution to spot obvious imbalances (overstocked vs understocked nodes)

Step 6: Feature Engineering / Parameter Estimation

- Calculate demand statistics per SKU-location: mean demand, standard deviation of demand over the lead time period
- Calculate lead time statistics: mean and standard deviation of lead times from each supply source
- Classify SKUs using ABC-XYZ analysis (ABC = value/volume importance, XYZ = demand variability) to determine which optimization approach and service level target applies to each segment
- Determine which network nodes can act as backup/pooling sources for each other (risk pooling potential)

Step 7: Model Building

- Calculate **safety stock** using standard formulas incorporating both demand and lead time variability (e.g., the classic safety stock formula: $Z\text{-score} \times \sqrt{(\text{lead time} \times \text{demand variance} + \text{demand}^2 \times \text{lead time variance})}$)
- Apply **multi-echelon inventory optimization (MEIO)** techniques, which jointly optimize stock levels across all network tiers simultaneously rather than each location independently — this captures **risk pooling** benefits (e.g., a central DC can serve as backup for multiple regional DCs, reducing total safety stock needed network-wide)
- Use simulation (Monte Carlo) to model network performance under different inventory policies, since analytical formulas can become intractable for complex multi-echelon networks
- Consider **service-level differentiation** — setting different target service levels for different SKU segments based on their strategic importance and margin

Step 8: Model Validation

- Backtest the proposed inventory policy against historical demand and lead time data — would it have prevented past stockouts while not over-investing in inventory?

- Simulate the policy under various demand scenarios (including extreme/peak scenarios) to test robustness
- Validate with supply chain planners — do the recommended stock levels align with practical constraints (minimum order quantities, storage capacity, supplier minimum lead times)?

Step 9: Business Case & Stakeholder Review

- Quantify the expected impact: projected inventory reduction, projected service level improvement, and financial impact (reduced carrying costs, reduced stockout-related lost sales)
- Present trade-off curves (e.g., service level vs inventory investment) so leadership can choose their preferred operating point
- Get sign-off from Supply Chain, Finance, and Operations leadership before implementation

Step 10: Deployment

- Integrate the new safety stock/reorder point parameters into the inventory planning system (ERP or specialized planning software) so replenishment decisions are automated based on the optimized parameters
- Roll out in phases — start with a pilot category or region before full network deployment
- Train planners on the new methodology and provide override capabilities for exceptional circumstances

Step 11: Monitoring & Maintenance

- Track actual service levels and inventory levels against targets on an ongoing basis
- Monitor forecast accuracy and lead time variability for drift, recalculating safety stock parameters periodically (e.g., quarterly) as underlying demand/supply patterns evolve
- Set up alerts for significant deviations from expected inventory or service level performance

Step 12: Reporting & Stakeholder Communication

- Present ongoing performance (inventory reduction achieved, service level trends) to supply chain leadership

- Provide insights back to demand planning and procurement teams on which SKUs/locations continue to be challenging, informing further process improvements

C. This Lifecycle Applies (with Variations) to Other Supply Chain Use Cases

Use Case	Key Lifecycle Differences
Transportation/Route Optimization	Shifts to operations research techniques (vehicle routing problem) rather than statistical inventory formulas; focuses on cost/time minimization subject to vehicle capacity and delivery window constraints
Network Design (Warehouse/DC Location Planning)	Longer planning horizon (multi-year), uses facility location optimization models balancing transportation costs, facility costs, and service coverage
Supplier Risk Management	Focuses on risk scoring and scenario/stress-testing rather than optimization, often incorporating external data (financial health, geopolitical risk)
Demand Forecasting	Shares strong overlap with the Retail & E-commerce domain's forecasting lifecycle, but supply chain forecasting often works at coarser SKU-family or category level for upstream planning

03. Key Metrics & KPIs

Inventory Metrics

Metric	Meaning
Inventory Turnover Ratio	How many times inventory is sold/used and replaced over a period
Days of Inventory on Hand (DOH)	Average number of days inventory would last at current usage rate
Fill Rate/Service Level	% of demand met immediately from available stock
Stockout Rate	% of time/instances where demand couldn't be met due to unavailable stock

Metric	Meaning
Inventory Carrying Cost	Total cost of holding inventory (storage, insurance, capital cost, obsolescence)
Excess/Obsolete Inventory %	% of inventory that is slow-moving or no longer sellable

Transportation & Logistics Metrics

Metric	Meaning
On-Time Delivery Rate (OTD)	% of shipments delivered by the promised date
On-Time In-Full (OTIF)	% of orders delivered both on time and with the complete quantity ordered
Freight Cost per Unit/per Mile	Transportation cost efficiency measure
Truck/Container Fill Rate	% utilization of available transportation capacity
Transit Time	Average time for goods to move between two points in the network
Perfect Order Rate	% of orders delivered on time, complete, damage-free, and with accurate documentation

Procurement & Supplier Metrics

Metric	Meaning
Supplier On-Time Delivery Rate	% of supplier deliveries received on schedule
Supplier Quality/Defect Rate	% of received goods that fail quality inspection
Purchase Price Variance (PPV)	Difference between planned and actual purchase cost
Supplier Lead Time	Average time from purchase order to receipt of goods

Warehouse & Fulfillment Metrics

Metric	Meaning
Order Cycle Time	Time from order placement to shipment/delivery

Metric	Meaning
Picking Accuracy	% of warehouse picks completed without errors
Warehouse Utilization	% of available storage capacity being used
Cost per Order Fulfilled	Total fulfillment cost divided by number of orders processed

Planning & Forecasting Metrics

Metric	Meaning
Forecast Accuracy (MAPE/WMAPE)	How close demand forecasts are to actual demand
Forecast Bias	Systematic tendency to over- or under-forecast
Bullwhip Effect Measure	Amplification of demand variability as it moves upstream through the supply chain

Overall Supply Chain Performance

Metric	Meaning
Total Supply Chain Cost as % of Revenue	Aggregate efficiency measure of the entire supply chain
Cash-to-Cash Cycle Time	Time between paying suppliers and receiving payment from customers
Supply Chain Resilience/Risk Score	Composite measure of exposure to disruption across the network

04. Common Datasets & Data Sources

Internal Data Sources

Source System	Typical Data
ERP Systems (SAP, Oracle)	Purchase orders, sales orders, inventory transactions, financial data

Source System	Typical Data
Warehouse Management System (WMS)	Stock levels, picking/packing data, warehouse location details
Transportation Management System (TMS)	Shipment records, carrier performance, freight costs, routing data
Demand Planning Systems (SAP IBP, Kinaxis, o9)	Historical demand, forecasts, planning parameters
Supplier Relationship Management (SRM) Systems	Supplier contracts, performance scorecards, quality records

External Data Sources

Source	Typical Data
GPS/Telematics Providers	Real-time vehicle location, transit time, driver behavior data
Weather Data Providers	Weather forecasts affecting transportation routes and demand
Port/Customs Authorities	Shipping schedules, customs clearance status, port congestion data
Freight Rate Indices (Freightos, Drewry)	Market freight rate benchmarks for ocean/air/trucking
Supplier Financial/Risk Data Providers (D&B, Moody's)	Supplier financial health and risk ratings

Typical Fields/Attributes an Analyst Works With

Demand/Order Data:

- Order ID, SKU, quantity, order date, requested delivery date, actual delivery date, customer/location

Inventory Data:

- SKU, location/node, on-hand quantity, in-transit quantity, safety stock level, reorder point

Transportation Data:

- Shipment ID, origin, destination, carrier, mode (truck/rail/ocean/air), departure/arrival time, freight cost, weight/volume

Supplier Data:

- Supplier ID, PO number, lead time, quantity ordered/received, quality inspection results, price

Data Characteristics Specific to Supply Chain & Logistics

- **Multi-echelon, networked structure** — data must be understood in the context of how nodes connect and depend on each other, not just in isolation
 - **High granularity and volume** — SKU × location × day-level data across large networks generates massive datasets
 - **Lead time and demand variability are both critical** — unlike simpler forecasting problems, supply chain analytics must jointly model uncertainty on both the supply and demand sides
 - **Data fragmentation across systems and partners** — data often lives across internal systems (ERP, WMS, TMS) and external partners (carriers, suppliers, 3PLs), requiring significant integration effort
 - **Constrained by real-world physical/operational limits** — models must respect capacity constraints (truck capacity, warehouse space, minimum order quantities) that don't exist in purely digital domains
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05. Tools & Technologies

Programming Languages

- **Python** – Pandas, NumPy, Scikit-learn, PuLP/OR-Tools (for optimization), SciPy for statistical calculations
- **R** – used for statistical forecasting and some optimization work
- **SQL** – for querying ERP/WMS/TMS data warehouses

Supply Chain Planning Software

- **SAP Integrated Business Planning (IBP)** – enterprise-grade demand/supply planning platform

- **Kinaxis RapidResponse** – widely used concurrent planning and supply chain control tower platform
- **o9 Solutions, Blue Yonder (formerly JDA)** – leading integrated business planning platforms
- **Llamasoft/Coupa Supply Chain Design** – network design and optimization software

Transportation & Logistics Software

- **Transportation Management Systems (TMS)**: Oracle TMS, SAP TM, MercuryGate
- **Route Optimization Tools**: OptimoRoute, Route4Me, or custom solutions built on Google OR-Tools

Optimization Libraries

- **Google OR-Tools** – open-source library for solving vehicle routing, scheduling, and network flow optimization problems
- **PuLP, Gurobi, CPLEX** – linear/mixed-integer programming solvers used for network optimization and inventory problems

Forecasting Tools

- **Prophet** – business-friendly time-series forecasting
- **Statsmodels, ARIMA-based tools** – classical statistical forecasting
- **Demand planning module forecasting engines** (within Kinaxis, SAP IBP, o9)

Simulation Tools

- **AnyLogic, Simio, Arena** – discrete event and agent-based simulation for supply chain network modeling
- **SimPy (Python)** – open-source discrete event simulation library

Visualization & BI Tools

- **Tableau, Power BI** – for supply chain KPI dashboards
- **Control tower dashboards** (within Kinaxis, o9, or custom-built) – for real-time network visibility

Cloud & Big Data Infrastructure

- **AWS, Azure, GCP** – for scalable data processing and ML deployment

- **Snowflake, Databricks** – for consolidating data across ERP/WMS/TMS systems
 - **Apache Airflow** – for orchestrating data pipelines across multiple supply chain systems
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06. Real-World Use Cases

Demand & Supply Planning

- **Demand Forecasting** – predicting future demand at SKU/category/location level to drive procurement and production planning
- **Sales & Operations Planning (S&OP) Support** – reconciling demand forecasts, supply capacity, and financial plans in cross-functional planning cycles
- **New Product Introduction (NPI) Forecasting** – forecasting demand for products with no historical sales data

Inventory Management

- **Multi-Echelon Inventory Optimization** – as detailed in the lifecycle example above
- **Safety Stock Optimization** – calculating optimal buffer stock levels balancing service level and carrying cost
- **Slow-Moving/Obsolete Inventory Analysis** – identifying and managing excess inventory risk

Transportation & Distribution

- **Vehicle Routing Optimization** – determining optimal delivery routes to minimize cost/time while meeting delivery windows
- **Freight Cost Optimization** – analyzing carrier/mode selection to minimize total transportation spend
- **Network Design/Distribution Center Location Planning** – determining optimal number and location of warehouses/DCs to balance cost and service coverage

Procurement & Supplier Management

- **Supplier Risk Scoring** – assessing and monitoring supplier financial health, geopolitical exposure, and performance risk
- **Should-Cost Modeling** – estimating the "true" fair cost of a purchased component to support supplier negotiations

- **Supplier Performance Scorecarding** – tracking on-time delivery, quality, and responsiveness across the supplier base

Warehouse & Fulfillment

- **Warehouse Slotting Optimization** – determining optimal storage locations for SKUs to minimize picking time
- **Labor Planning/Forecasting** – predicting warehouse staffing needs based on expected order volume
- **Order Fulfillment Optimization** – determining which warehouse/DC should fulfill each order to minimize cost and delivery time

Risk & Resilience

- **Supply Chain Risk/Disruption Simulation** – modeling the impact of potential disruptions (supplier failure, port closure, natural disaster) on the network
- **Scenario Planning** – stress-testing the supply chain against various demand and supply shock scenarios

Example Interview-Style Problem Statements

- *"How would you determine the optimal safety stock level for a SKU given variable demand and supplier lead times?"*
- *"Design a vehicle routing solution to minimize delivery costs for 50 daily delivery stops with time-window constraints."*
- *"Our on-time delivery rate has dropped 10% over the last quarter — how would you investigate the root cause?"*
- *"How would you decide whether to open a new regional distribution center given current demand and shipping cost data?"*

07. ML & Statistical Techniques

Demand Forecasting Techniques

- **ARIMA/SARIMA, Exponential Smoothing (Holt-Winters)** – classical time-series methods for stable demand patterns
- **Gradient Boosting (XGBoost, LightGBM)** – for demand forecasting incorporating many external features (promotions, holidays, pricing)

- **Hierarchical/Reconciled Forecasting** – ensuring forecasts are consistent across SKU, category, and network levels

Inventory Optimization Techniques

- **Safety Stock Formulas (based on Normal distribution assumptions)** – foundational statistical approach incorporating demand and lead time variability
- **Multi-Echelon Inventory Optimization (MEIO)** – advanced techniques (e.g., guaranteed service model, stochastic-service model) that jointly optimize stock across network tiers
- **Newsvendor Model** – classical operations research model for single-period inventory decisions under uncertain demand
- **Monte Carlo Simulation** – for modeling complex, non-analytically-tractable multi-echelon network behavior

Transportation & Routing Optimization

- **Vehicle Routing Problem (VRP) algorithms** – exact and heuristic methods (nearest neighbor, genetic algorithms, simulated annealing) for route optimization
- **Linear/Mixed-Integer Programming** – for network flow, facility location, and transportation cost optimization
- **Traveling Salesman Problem (TSP) variants** – foundational routing optimization problem, extended with real-world constraints in VRP

Network Design & Facility Location

- **Facility Location Optimization Models** – mixed-integer programming models balancing facility costs, transportation costs, and service coverage
- **Center of Gravity Method** – simple heuristic for initial warehouse location analysis based on demand-weighted geography

Risk & Classification Techniques

- **ABC-XYZ Analysis** – classification technique combining value/volume importance (ABC) with demand variability (XYZ) to segment inventory management approaches
- **Logistic Regression, Random Forest** – for supplier risk scoring and disruption prediction
- **Survival Analysis** – for modeling time-to-disruption or supplier failure risk

Statistical Process & Quality Control (Shared with Manufacturing)

- **Control Charts** – for monitoring supply chain process metrics (lead time, quality) over time
- **Root Cause Analysis Techniques** – for investigating service level failures or cost overruns

Simulation & Scenario Analysis

- **Discrete Event Simulation** – modeling warehouse operations, order fulfillment flows, and network behavior under various scenarios
 - **Agent-Based Modeling** – simulating individual entities (trucks, orders, suppliers) and their interactions for complex system behavior analysis
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08. Resources

Public Datasets for Practice

- **Kaggle: "DataCo Smart Supply Chain Dataset"** – comprehensive supply chain dataset covering orders, shipping, and customer data
- **Kaggle: "Supply Chain Shipment Pricing Data"** – for freight cost analysis practice
- **M5 Forecasting Competition Dataset (Kaggle)** – large-scale hierarchical demand forecasting dataset (shared reference with Retail domain)
- **Kaggle: "Logistics and Supply Chain Dataset"** (various) – multiple practice datasets for inventory and logistics analysis

Recommended Courses

- Supply Chain Management Specialization (Coursera – Rutgers University)
- Supply Chain Analytics courses (Coursera – MIT, various universities)
- Operations Research/Optimization courses (Coursera/edX)
- Inventory Management and Demand Planning courses (Coursera/LinkedIn Learning)

Books

- *"Designing and Managing the Supply Chain"* – Simchi-Levi, Kaminsky, Simchi-Levi (a foundational textbook)
- *"Inventory Management and Production Planning and Scheduling"* – Edward Silver (classic inventory theory reference)

- *"Supply Chain Analytics"* – various practitioner-focused references combining theory with applied data science
- *"The Goal"* – Eliyahu Goldratt (foundational, narrative-style introduction to operations/theory of constraints thinking)

Communities & Blogs

- CSCMP (Council of Supply Chain Management Professionals) – industry association and resources
- Supply Chain Digital, SupplyChainBrain – industry news and case studies
- Kaggle competition discussions on supply chain/logistics datasets
- LinkedIn groups: "Supply Chain Analytics", "Operations Research and Supply Chain"

Standards & Reference Material

- SCOR Model documentation (Association for Supply Chain Management, ASCM) – standardized process/metric framework
- APICS/ASCM certification materials (CPIM, CSCP) – widely recognized supply chain professional certifications

GitHub Repositories for Reference

- Search GitHub for: vehicle-routing-problem, inventory-optimization, supply-chain-network-design, demand-forecasting-hierarchical for open-source implementation examples

This completes the Supply Chain & Logistics domain notes.